Noise and Human Capital

Armando Meier Felix Schulz

Noise pollution is commonplace, but its consequences for human capital formation are less studied than those of air pollution or heat. Our goal is to provide first causal, long-run evidence on the impact of noise externalities on human capital formation. For an overview of central papers, see Table 1.

1 Previous evidence

The most relevant papers: Identification of noise impacts. There are two main issues in identifying the causal impact of noise: residential sorting and air pollution concurring with noise pollution. A few key papers convincingly address these issues using either wind direction and noise from non-polluting, electric trains (Greenhill 2026), sound barriers construction around roads (Moretti and Wheeler 2025), or noise proofing of houses (Lindgren 2021). These papers study the impact on health at birth (Greenhill 2026), hospitalizations because of cardiovascular disease (Lindgren 2021), and property prices (Moretti and Wheeler 2025), but not on human capital formation. Studying impacts on human capital while addressing the two main identification issues is our goal.

Limited evidence on productivity and cognition. Evidence on cognition and productivity highlights the potential of noise also impacting broader human capital formation. In a field experiment, Dean (2024) shows experimentally that a 7dB increase in workplace noise reduces textile-worker output by about 3 percent and reduces an index of attention, memory, inhibition and further measures capturing cognition. The most convincing design studying impacts on cognition among children is that of Hygge et al. (2002), who exploit the Munich airport relocation as a natural experiment. Testing schoolchildren matched on socioeconomic characteristics before and after the airport move, they find that children newly exposed to aircraft noise show impaired reading and long-term memory, while those who lost exposure show improvements. Other papers find consistent cross-sectional associations of higher noise with worse memory and reading skills (Clark et al. 2010; Haines et al. 2001; Klatte et al. 2017; Lercher et al. 2003). Many other papers also measure the link between noise and educational outcomes in the cross-section, but cannot address the two main identification issues.

Noise likely deteriorates sleep, well-being, and health. Existing literature establishes a plausible link between noise and health indicators. Apart from distraction by noise, that is, a direct impact on cognition and learning discussed above, this is likely one of the most important mechanisms.

Two economics papers seem particularly important here: First, Greenhill (2026, working paper) constructs a new noise measure by using night-time seismometer data capturing electric train traffic in combination with wind direction. Greenhill (2026) finds that nighttime rail-noise during pregnancy reduces birthweight and an index of newborn health. Second, Lindgren (2021, working paper) finds that improved house insulation against noise reduced hospitalizations for cardiovascular disease.

Beyond these two most closely related papers, Boes et al. (2013) exploit two unexpected flight-regulation changes in Zurich and find that noise increases sleeping problems and headaches. The causal estimates are larger than cross-sectional estimates because of sorting (Boes et al. 2013). Further, Beghelli et al. (2023) exploit an unannounced reallocation of early-morning landings at Heathrow and find that prescription spending for sleep disorders, anxiety, depression as well as respiratory disease falls when noise drops (see also Wang et al. 2022). And, finally, Krekel and Zerrahn (2017) and Krekel et al. (2023) use a differences-in-differences design, finding lower well-being close to wind turbines but no health effects (see also, Zou 2017). A broader public health and economics literature points in the same direction but are either laboratory experiments or less well identified, relying more on self-reported exposure and on cross-sectional variation (see, e.g., Van Praag and Baarsma 2005; Basner and McGuire 2018; Basner 2008; Griefahn et al. 2006; Bodin et al. 2015; Trimmel et al. 2012).

2 Data and Identification

Table 2 provides an overview of potential data sources.

Variation in Noise Exposure: Barriers and Dwelling Insulation

We identify exposure to noise pollution using the staggered rollout of mitigation technology: Trafikverket, the Swedish transportation authority, provides dwellings and neighborhoods with sound insulation and noise barriers, respectively. The BoViT program covers all provided insulation and barriers, including the exact dwellings treated and the date of provision. Data on dwelling insulation are, however, not publicly available and must be requested from Trafikverket. A smaller, publicly available subset covers only noise barriers, and construction dates are missing for many of these.

Supplementary, publicly available data on dwelling, road, and rail positions let us determine which dwellings are located close enough to a road or railway to plausibly be eligible for treatment–i.e., to construct a sample of at-risk dwellings used in the staggered rollout design–and we can additionally request modeled background noise exposure (for all of Sweden) and historical traffic density.

Outcomes: Health and Educational Attainment

We plan to study health and education outcomes based on administrative data. For instance, in adolescence, National Patient Register diagnosis codes would let us study mental health, sleep disorders, and insomnia. At birth, the Medical Birth Register would provide birthweight and other neonatal health measures for children exposed to noise in utero. In adulthood, register data would let us study diagnosed cardiovascular diseases including hypertension (similar to Lindgren 2021).

Alongside health, students' learning outcomes will be shaped by noise pollution they are exposed to at school and at home. With publicly available school-level data, we can identify the effects of noise barriers next to school buildings. Individual-level test scores from Skolverket, matched via the geocoded population register (RTB), would let us identify the effects of home noise insulation. Finally, access to the labor-market register would let us identify longer-term outcomes.

Potential Identification Strategies

- **Noise barriers (similar to Moretti and Wheeler 2025):** exploit the staggered construction of road/rail noise barriers, comparing dwellings and schools just inside vs. just outside the newly shielded area–separately for home exposure (residential DiD) and school exposure (e.g., school-level DiD using national test data), following Moretti and Wheeler (2025).
- **Sound insulation (similar to Lindgren 2021):** exploit the staggered timing of BoViT insulation as in Lindgren (2021), but compare treated dwellings to not-yet-treated eligible dwellings rather than to never-treated ones.

For either strategy, we may use the timing of insulation or barrier construction relative to a child's birth or school-entry date to construct historical exposure measures to isolate in-utero and early-childhood exposure windows. Across these designs, we can account for dwelling/school fixed effects, cohort or year-of-birth fixed effects, and family socioeconomic status from the population register.

A key issue is that property values move with noise levels (Allroggen et al. 2025; Moretti and Wheeler 2025). But this seems potentially addressable as we would know sorting based on individual-level data and depending on exposure timing (e.g., via sibling fixed effects). Also, barriers built around schools would likely lead to much less sorting and/or wealth effects than closeby where people live.

Literature overview

Table 1. Selected studies on noise pollution, well-being, health, and human capital (more in text). Table extends overview in Nyberg (2025).

Study	Research Question	Identification	Data & Sample	Main Result	Country	Period
Greenhill (2026)	Does nighttime noise harm infant health?	Quasi-exp. using idiosyncratic wind-driven rail-noise variation (share of pregnancy spent downwind)	Seismometer-based noise + confidential birth-certificate microdata	+2dB nighttime noise → -4% SD infant health index	USA (California)	Births: 1999–2021
Lindgren (2021)	Does a dwelling noise-mitigation program reduce cardiovascular disease?	DiD around a nationwide program	Hospitalization/diagnosis registers; 10,113 indiv. who stayed at the same address throughout	-10% cardiovascular disease risk after 7 years (-18–19 cases/1,000), two thirds via hypertension	Sweden	Residents: 1998–2008; mitigation: 1999–2017; outcomes: 2003–2017
Moretti and Wheeler (2025)	Are noise externalities capitalized into property prices, and how large is the aggregate cost?	DiD/event study around FDOT noise-barrier construction; hedonic house-price model	FL barrier inventory, 1,143 barriers; CoreLogic transactions, 596,419 home sales in the estimation sample	Noise barriers raise nearby property values by 6.8% within 100m; implies $110bn aggregate U.S. cost, $974/vehicle efficient tax	USA (Florida)	Barriers: 1988–2022; transactions: 1990–2022
Dean (2024)	Does workplace noise reduce productivity and cognition, and do workers realize it?	Two randomized field experiments: randomized engine-noise exposure during a sewing task, plus a willingness-to-pay-for-quiet elicitation	Textile-training course; sewing task (N=128)	+7dB noise → -3% output, impaired cognition; no WTP premium for quiet (workers unaware of the cost)	Kenya (Nairobi)	Not stated
Hygge et al. (2002)	Does a change in aircraft-noise exposure affect children's cognition?	Prospective natural experiment around the Munich airport relocation; 3 testing waves (6mo before, 1yr & 2yr after)	326 children, 8–12y, matched noise/no-noise groups at old & new sites	Reading and long-term memory impaired at newly-exposed site, improved at newly-quiet site	Germany (Munich)	1991–1994
Clark et al. (2010)	Does aircraft/road noise affect children's cognition, and is the relationship exposure–response?	Multilevel cross-sectional; cued recall, recognition & prospective-memory tests	2,844 children, 8;10–12;10y, matched schools around 3 airports	Aircraft noise → worse recognition memory (linear); road noise → better cued recall (linear, unexpected direction)	UK, Netherlands, Spain	early 2000s
Haines et al. (2001)	Does chronic aircraft noise impair children's reading and mental health?	Quasi-exp. matched-school design; group-administered cognitive/mental-health tests + salivary cortisol subsample	4 high-/4 low-noise schools; 340 children, 8–11y (169 high-/171 low-noise)	Worse reading comprehension; no mental-health or cortisol effect	UK (Heathrow)	1991
Klatte et al. (2017)	Does aircraft noise lower reading performance at moderate exposure levels?	Multilevel cross-sectional, continuous exposure–response design (not just hi/lo groups); reading & quality-of-life tests + teacher reports	1,243 2nd graders, 29 schools; teacher reports (N=84)	Even moderate noise (<60dB) lowers reading performance, raises annoyance; no effect on verbal precursors	Germany (Frankfurt)	Noise: May 2011–Apr 2012; testing: spring (year not stated)

Data sources

Table 2. Data sources for exposure, outcome, and auxiliary variables.

Role	Dataset	Source	Level of Detail	Period	Open Access?	Access
Exposure	BoViT project data, incl. dwelling-level noise insulation[1]	Trafikverket	Dwelling-level, geocoded	TBD – confirm with Trafikverket	No	Request from Trafikverket
Exposure	Sound barriers – roads	Trafikverket	Barrier-segment level, geocoded (road network); 2,172 segments in the current extract	1950–2025 (51% unrecorded)	Yes	Lastkajen entry 630
Exposure	Sound barriers – railway	Trafikverket	Barrier-segment level, geocoded (rail network); 1,829 segments in the current extract	1989–2025 (27% unrecorded)	Yes	Lastkajen entry 991
Outcome	National test & final-grade results, school-level aggregates[2]	Skolverket	School-unit level (not individual, not geocoded itself); grade-point average & national-test pass-rate measures	1998–2019 + 2022/23–2025/26 (gap ∼2020–2022)	Yes	1998–2019: archived SIRIS exports on Skolverket's S3 (via jplusplus/skolstatistik); 2022/23–: Skolverkets statistikdatabas (PxWeb API)
Outcome	National test & final-grade results, individual-level microdata	Skolverket	Individual student-level microdata	TBD	No	Microdata request via Skolverket
Outcome	Labor-market register: earnings, employment status	Statistics Sweden, SCB	Individual-level microdata	1990–present (LISA, confirm exact span)	No	Microdata service (MONA)
Auxiliary	School location register (Skolenhetsregistret)	Skolverket	School-level, geocoded (address/coordinates)	Current only	Yes	Open API (skolverket.se)
Auxiliary	Geocoded population register: student-dwelling linkage, RTB	Statistics Sweden, SCB	Individual-level, geocoded dwelling address	TBD	No	Microdata service (MONA)

[1] Previously used by Lindgren (2021).

[2] School-level statistics were suppressed from Sept. 2020: a Dec. 2019 Kammarrätten i Göteborg ruling that independent-school throughput/grade data could be a trade secret led Skolverket to stop publishing any school-unit-level statistics, public or independent. A temporary law (July 2021, initially through June 2023) restored publication retroactively; school-unit-level data is confirmed live today via the PxWeb API. Since spring 2025, small-cell suppression applies specifically to *staff* counts (<3 FTE at a school); no equivalent grade/test-result small-cell rule was found.

References

Allroggen, Florian et al. (Oct. 2025). *Planes Overhead: How Airplane Noise Impacts Home Values*. w34431. Cambridge, MA: National Bureau of Economic Research, w34431. DOI: 10.3386/w34431.

Basner, M. (June 2008). "Nocturnal Aircraft Noise Exposure Increases Objectively Assessed Daytime Sleepiness". In: *Somnologie - Schlafforschung und Schlafmedizin* 12.2, pp. 110–117. ISSN: 1432-9123, 1439-054X. DOI: 10.1007/s11818-008-0338-8.

Basner, Mathias and Sarah McGuire (Mar. 14, 2018). "WHO Environmental Noise Guidelines for the European Region: A Systematic Review on Environmental Noise and Effects on Sleep". In: *International Journal of Environmental Research and Public Health* 15.3, p. 519. ISSN: 1660-4601. DOI: 10.3390/ijerph15030519.

Beghelli, Silvia, Augustin De Coulon, and Mary O'Mahony (Oct. 2023). "Health Benefits of Reducing Aircraft Pollution: Evidence from Changes in Flight Paths". In: *Journal of Population Economics* 36.4, pp. 2581–2607. ISSN: 0933-1433, 1432-1475. DOI: 10.1007/s00148-023-00964-y.

Bodin, Theo et al. (Jan. 29, 2015). "Annoyance, Sleep and Concentration Problems Due to Combined Traffic Noise and the Benefit of Quiet Side". In: *International Journal of Environmental Research and Public Health* 12.2, pp. 1612–1628. ISSN: 1660-4601. DOI: 10.3390/ijerph120201612.

Boes, Stefan, Stephan Nüesch, and Steven Stillman (Sept. 2013). "AIRCRAFT NOISE, HEALTH, AND RESIDENTIAL SORTING: EVIDENCE FROM TWO QUASI-EXPERIMENTS". In: *Health Economics* 22.9, pp. 1037–1051. ISSN: 1057-9230, 1099-1050. DOI: 10.1002/hec.2948.

Clark, Charlotte et al. (2010). "The Effects of Road Traffic and Aircraft Noise Exposure on Children's Episodic Memory: The RANCH Project". In: *Noise and Health* 12.49, p. 244. ISSN: 1463-1741. DOI: 10.4103/1463-1741.70503.

Dean, Joshua T. (Oct. 1, 2024). "Noise, Cognitive Function, and Worker Productivity". In: *American Economic Journal: Applied Economics* 16.4, pp. 322–360. ISSN: 1945-7782, 1945-7790. DOI: 10.1257/app.20220532.

Greenhill, Simon (2026). *Noise Pollution and Infant Health*. URL: https://simondgreenhill.github.io/Greenhill_Noise.pdf.

Griefahn, Barbara, Anke Marks, and Sibylle Robens (Aug. 2006). "Noise Emitted from Road, Rail and Air Traffic and Their Effects on Sleep". In: *Journal of Sound and Vibration* 295.1–2, pp. 129–140. ISSN: 0022460X. DOI: 10.1016/j.jsv.2005.12.052.

Haines, M. M. et al. (Feb. 2001). "Chronic Aircraft Noise Exposure, Stress Responses, Mental Health and Cognitive Performance in School Children". In: *Psychological Medicine* 31.2, pp. 265–277. ISSN: 0033-2917, 1469-8978. DOI: 10.1017/S0033291701003282.

Hygge, Staffan, Gary W. Evans, and Monika Bullinger (Sept. 2002). "A Prospective Study of Some Effects of Aircraft Noise on Cognitive Performance in Schoolchildren". In: *Psychological Science* 13.5, pp. 469–474. ISSN: 0956-7976, 1467-9280. DOI: 10.1111/1467-9280.00483.

Klatte, Maria et al. (May 2017). "Effects of Aircraft Noise on Reading and Quality of Life in Primary School Children in Germany: Results From the NORAH Study". In: *Environment and Behavior* 49.4, pp. 390–424. ISSN: 0013-9165, 1552-390X. DOI: 10.1177/0013916516642580.

Krekel, Christian, Johannes Rode, and Alexander Roth (2023). *Do Wind Turbines Have Adverse Health Impacts?* IZA Discussion Papers 16505. Bonn: Institute of Labor Economics (IZA). HDL: 10419/282632. URL: https://hdl.handle.net/10419/282632.

Krekel, Christian and Alexander Zerrahn (Mar. 2017). "Does the Presence of Wind Turbines Have Negative Externalities for People in Their Surroundings? Evidence from Well-Being Data". In: *Journal of Environmental Economics and Management* 82, pp. 221–238. ISSN: 00950696. DOI: 10.1016/j.jeem.2016.11.009.

Lercher, Peter, Gary W. Evans, and Markus Meis (Nov. 2003). "Ambient Noise and Cognitive Processes among Primary Schoolchildren". In: *Environment and Behavior* 35.6, pp. 725–735. ISSN: 0013-9165, 1552-390X. DOI: 10.1177/0013916503256260.

Lindgren, Samuel (Oct. 2021). *A Sound Environment: Health Effects of Traffic Noise Mitigation*. Working Papers 2021:10. Swedish National Road & Transport Research Institute (VTI). URL: https://ideas.repec.org/p/hhs/vtiwps/2021_010.html.

Moretti, Enrico and Harrison Wheeler (Sept. 2025). *The Traffic Noise Externality: Costs, Incidence and Policy Implications*. w34298. Cambridge, MA: National Bureau of Economic Research, w34298. DOI: 10.3386/w34298.

Nyberg, Erik (Nov. 2025). *Air and Noise Pollution: A State-of-the-Art Review of Causal Health Outcomes*. Working Papers 2025:6. Swedish National Road & Transport Research Institute (VTI). URL: https://ideas.repec.org/p/hhs/vtiwps/2025_006.html.

Trimmel, Michael et al. (Nov. 2012). "Effects of Low Intensity Noise from Aircraft or from Neighbourhood on Cognitive Learning and Electrophysiological Stress Responses". In: *International Journal of Hygiene and Environmental Health* 215.6, pp. 547–554. ISSN: 14384639. DOI: 10.1016/j.ijheh.2011.12.007.

Van Praag, Bernard M. S. and Barbara E. Baarsma (Jan. 1, 2005). "Using Happiness Surveys to Value Intangibles: The Case of Airport Noise". In: *The Economic Journal* 115.500, pp. 224–246. ISSN: 0013-0133, 1468-0297. DOI: 10.1111/j.1468-0297.2004.00967.x.

Wang, Scarlett Sijia et al. (May 2022). "Impact of Aeroplane Noise on Mental and Physical Health: A Quasi-Experimental Analysis". In: *BMJ Open* 12.5, e057209. ISSN: 2044-6055, 2044-6055. DOI: 10.1136/bmjopen-2021-057209.

Zou, Eric Yongchen (2017). *Wind Turbine Syndrome: The Impact of Wind Farms on Suicide*. URL: https://api.semanticscholar.org/CorpusID:160015504.